

5.4 Electrochemistry Calculations & Applications (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.4 Electrochemistry Calculations & Applications (A Level Only)
Difficulty	Medium

Time allowed: 70

Score: /53

Percentage: /100

Question 1a

This question is about electrolysis cells.

State the relationship between the Faraday constant, F , the charge on an electron, e , and the Avogadro constant, L .

[1 mark]

Question 1b

If the charge on the electron, the A_r and the charge of copper ions are known, the Avogadro constant can be determined experimentally by passing a known current for a known time through a copper electrolysis cell and weighing the mass of copper deposited onto the cathode.

Draw a diagram of apparatus which is suitable for carrying out this practical.

Label the following:

- power supply with the + and - terminals identified
- anode
- cathode
- ammeter

State the composition of the electrolyte.

[4 marks]

Question 1c

The following are the results obtained from one such experiment.

current passed through the cell	= 0.375 A
time current was passed through cell	= 40.0 min
initial mass of copper cathode	= 49.637 g
final mass of copper cathode	= 49.936 g

Use the experimental results to calculate a value of the Avogadro constant, L , to 3 significant figures.

[electronic charge, $e = -1.60 \times 10^{-19}$ C]

[5 marks]

Question 1d

Table 2.1 shows a list of standard electrode potentials at 298 K.

Table 2.1

Electrode reaction	$E^\ominus / \text{V}$
$\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$	+0.80
$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	+1.07
$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$	+2.87
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.44
$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$	0.00
$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	-2.38
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightleftharpoons 4\text{OH}^-$	+0.40
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{SO}_2 + 2\text{H}_2\text{O}$	+0.17

Use the information in **Table 2.1** to determine the substances formed at the anode and cathode when the following substances are electrolysed.

compound	product at anode	product at cathode
AgF		
FeSO ₄		
MgBr ₂		

[5 marks]

Question 2a

An electrochemical cell is set up as shown in Fig. 2.1.

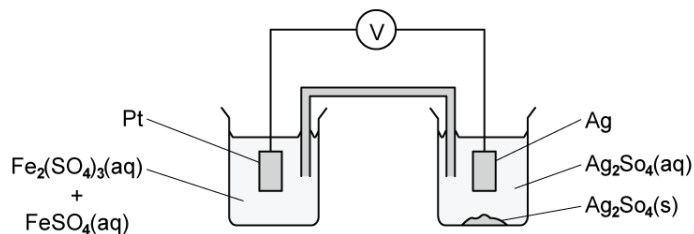


Fig. 2.1

Use the electrode potential list in Table 2.1 to calculate the value of $E^\ominus_{\text{cell}}$ under standard conditions, stating which electrode is the negative one.

Table 2.1

Electrode reaction	$E^\ominus / \text{V}$
$\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$	+0.80
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.44
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+0.77
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{SO}_2 + 2\text{H}_2\text{O}$	+0.17

$E^\ominus_{\text{cell}} = \dots\dots\dots$

negative electrode = $\dots\dots\dots$

[1 mark]

Question 2b

How would the actual E_{cell} of the above cell compare to the $E^\ominus_{\text{cell}}$ under standard conditions?

Explain your answer.

[2 marks]

Question 2c

How would the E_{cell} of the cell in Fig. 2.1 change, if at all, if a few cm^3 of concentrated $\text{Na}_2\text{SO}_4(\text{aq})$ were added to the following?

i)

The beaker containing $\text{Fe}^{3+}(\text{aq})$ and $\text{Fe}^{2+}(\text{aq})$.

[1]

ii)

The beaker containing $\text{Ag}_2\text{SO}_4(\text{aq})$.

[1]

iii)

Explain any changes in E_{cell} you have stated in (i) and (ii).

[2]

[4 marks]

Question 2d

Write an equation to show the reaction taking place in the electrochemical cell in Fig. 2.1.

[1 mark]

Question 3a

The standard electrode potentials for seven different redox systems are shown in **Table 3.1**.

Table 3.1

redox system	equation	$E^\ominus / \text{V}$
1	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$	0.00
2	$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Fe}^{2+}(\text{aq})$	+0.77
3	$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$	+0.34
4	$\text{Cl}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq})$	+1.36
5	$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}(\text{l})$	+1.23
6	$\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightleftharpoons \text{Al}(\text{s})$	-1.66
7	$\text{I}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{I}^-(\text{aq})$	+0.54

An electrochemical cell can be made based on redox systems **2** and **3**.

i)

Write the overall cell reaction.

[1]

ii)

Calculate the voltage of this cell.

[1]

[2 marks]

Question 3b

i)

Using redox systems **5**, **6** and **7 only** in **Table 3.1**, write the overall equations for **three** reactions that might be feasible.

[3]

ii)

Give **two** reasons why these reactions might not take place, even if they are feasible.

[2]

[5 marks]

Question 3c

Select from **Table 3.1**,

i)
an oxidising agent that oxidises $\text{Fe}^{2+}(\text{aq})$ to $\text{Fe}^{3+}(\text{aq})$,

[1]

ii)
a species that reduces $\text{Fe}^{3+}(\text{aq})$ to Fe^{2+} but does not reduce $\text{Cu}^{2+}(\text{aq})$ to $\text{Cu}(\text{s})$.

[1]

[2 marks]**Question 3d**

Calculate the electrode potential at 298 K of redox system **3** if the concentration of $\text{Cu}^{2+}(\text{aq})$ ions was $0.0002 \text{ mol dm}^{-3}$.

[3 marks]

Question 4a

Table 4.1 below contains some standard electrode potential data which you will need to answer the following questions.

Table 4.1

electrode reaction	E^θ / V
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$	+0.34
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Ni}(\text{s})$	-0.25
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Sn}(\text{s})$	-0.14
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	-0.45

Deduce the species from **Table 1** that is the weakest oxidising agent. Explain your choice.

[2 marks]

Question 4b

A cell is made by connecting two half-cells with a salt bridge.

A student produced a cell by using nickel in a solution of nickel chloride solution and another consisted of copper in a solution of copper sulfate solution.

Calculate the standard cell potential of this cell using the values given in **Table 4.1** in part (a).

[1 mark]

Question 4c

Calculate the standard Gibbs free energy change, and give the units, for the electrochemical cell in part (b) and state whether the reaction is feasible.

Show your working.

[Faraday constant, $F = 9.65 \times 10^4 \text{ C mol}^{-1}$]

[3 marks]

Question 4d

Two half-cells, involving species in **Table 4.1**, are connected together to give a cell with a standard cell potential = +0.31 V.

i)

Determine which two half equations are connected using the data from **Table 4.1**.

[1]

ii)

Suggest the half-equation for the reaction that occurs at the cathode.

[1]

[2 marks]

Question 5a

The standard electrode potentials in **Table 5.1** can be used to predict redox reactions.

Table 5.1

redox system	equation	$E^{\ominus} / \text{V}$
1	$\text{Ag}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Ag}(\text{s})$	+0.80
2	$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightleftharpoons \text{Cr}(\text{s})$	-0.74
3	$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Mg}(\text{s})$	-2.37

Using the information in **Table 5.1**, write equations for the reactions that are feasible.

[3 marks]

Question 5b

A student sets up a standard cell, using half-cells based on redox systems **2** and **3** at 298 K.

i)

Calculate the standard cell potential.

[1]

ii)

State the sign of the electrode in redox system **2** of the cell.

[1]

[2 marks]

Question 5c

The student diluted the solution in redox system **3** with distilled water.

Predict what would happen to the cell potential.

Explain your reasoning.

[3 marks]

Question 5d

The student set up the cell between redox systems **2** and **3** again. This time, they did not dilute the solution of redox system **3** but they used a solution of 0.01 mol dm^{-3} of Cr^{3+} ions for redox system **2**. The temperature remained at 298 K.

Use the Nernst equation below to predict what the new electrode potential, E , would be for redox system **2**.

$$E = E^\ominus + \frac{0.059}{z} \log_{10} \frac{[\text{oxidised species}]}{[\text{reduced species}]}$$

where z = the number of electrons transferred in the reaction.

[2 marks]



Model Answers: Medium

1a

a) The relationship between the Faraday constant, F , the charge on an electron, e , and the Avogadro constant, L is:

- $F = Le$

OR

$$L = F/e$$

OR

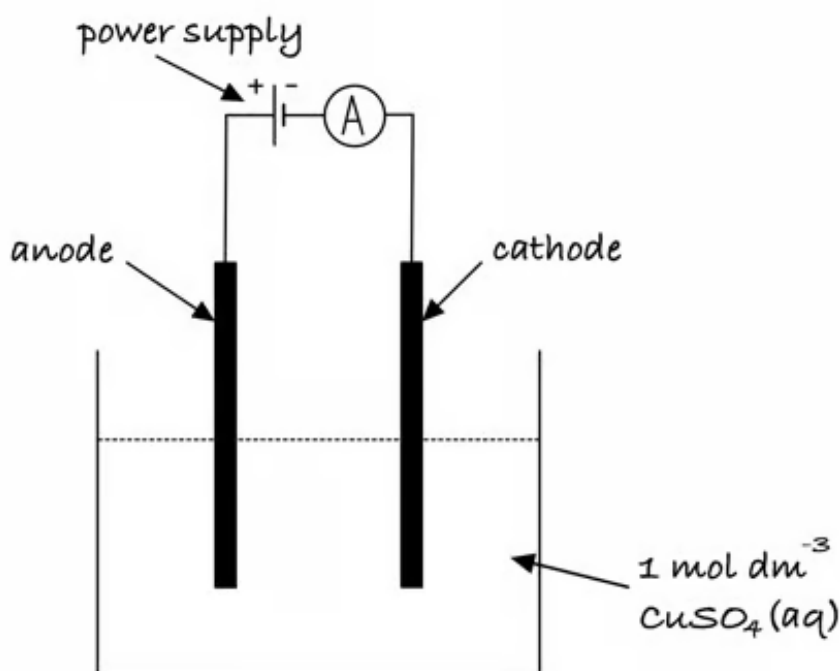
$$e = F/L; [1 \text{ mark}]$$

[Total: 1 mark]

- This is known as Faraday's Law which defines one faraday as the amount of electric charge carried by 1 mole of electrons or 1 mole of singly charged ions
- You need to know and be able to apply this relationship

1b

b) A labelled diagram of the apparatus which is suitable for carrying out this practical and the composition of the electrolyte is:



- Correct cell to include 2 electrodes and power supply circuit; [1 mark]
- Ammeter in series; [1 mark]
- Anode and cathode correctly labelled with the right polarity; [1 mark]
- $\text{CuSO}_4(\text{aq}) / \text{CuCl}_2(\text{aq}) / \text{Cu}^{2+}(\text{aq})$

AND

1 mol dm^{-3} ; [1 mark]

[Total: 4 marks]

- This circuit often contains a variable resistor so that the current can be adjusted to a suitable value which is then kept the same during the experiment, however, it was not necessary to include one to gain full marks
- Both electrodes will be made from copper – the cathode will increase in mass as copper is deposited on it and the anode will decrease in mass as copper goes into solution as copper ions
- Calculations based on this experiment usually use the decrease in mass of the anode as not all the copper formed will stick to the cathode properly so it would give a less accurate value of L

1c

c) A value of the Avogadro constant, L , to 3 significant figures is:

Method 1:

- Amount of charge passed, $Q = It = 0.375 \times 40.0 \times 60 = 900 \text{ C}$; [1 mark]
- $n(\text{Cu}) = (49.936 - 49.637)/63.5 = 4.71 \times 10^{-3} \text{ mol}$; [1 mark]
- Amount of charge to deposit 1 mol Cu = $900/4.71 \times 10^{-3} = 191082.8 \text{ C}$; [1 mark]
- Charge on 1 mol $e^- = 191082.8/2 = 95541.4 \text{ C}$; [1 mark]
- $L = \text{charge on 1 mol } e^- / \text{charge on 1 } e^- = 95541.4/1.60 \times 10^{-19} = 5.97 \times 10^{23} \text{ mol}^{-1}$; [1 mark]

Method 2:

- $n(\text{Cu}) = (49.936 - 49.637)/63.5 = 4.71 \times 10^{-3} \text{ mol}$; [1 mark]
- $n(e^-) \text{ required} = 4.71 \times 10^{-3} \times 2 = 9.42 \times 10^{-3} \text{ mol}$; [1 mark]
- Amount of charge passed, $Q = It = 0.375 \times 40.0 \times 60 = 900 \text{ C}$; [1 mark]
- Number of e^- passed = $900/1.60 \times 10^{-19} = 5.625 \times 10^{21}$; [1 mark]
- Number of e^- passed/ $n(e^-)$ required = $L = 5.625 \times 10^{21}/9.42 \times 10^{-3} = 5.97 \times 10^{23} \text{ mol}^{-1}$; [1 mark]

[Total: 5 marks]

- In method 1, if you used the calculator value for the number of moles of Cu to calculate the amount of charge to deposit 1 mol Cu, you would get a slightly different value of 191137.1 C
 - This would give the charge on 1 mol e^- to be 95568.6 C
 - This would still give the final value, to 3 significant figures, to be the same as the mark scheme
- In method 2, you can also give the first and second marking points to a greater number of significant figures which would still lead to the same final answer
- Either method is valid, so use the method that makes the most sense to you and spend time practicing this type of calculation - these are straightforward calculations if you know how
- **Remember:** 1 mol of copper ions will react with 2 moles of electrons to deposit one mole of copper
- Your final answer must be to 3 significant figures, you will not score the final mark if it is not

1d

d) The substances formed at each electrode are:

compound	product at anode	product at cathode

AgF	F ₂	Ag
FeSO ₄	O ₂	H ₂
MgBr ₂	Br ₂	H ₂

- 6 correct; [5 marks]
- 5 correct; [4 marks]
- 4 correct; [3 marks]
- 3 correct; [2 marks]
- 2 correct; [1 mark]

[Total: 5 marks]

- You would be awarded the marks if you gave the names instead of the symbols
- If the atomic symbol(e.g. Br or O) is used instead of the molecular formula then 1 mark is deducted
- **Remember:** In an aqueous solution, ions from water, H⁺ and OH⁻, are also present and can be discharged
- The **positively charged cation** with the most **positive** $E^\ominus$ will be discharged at the **cathode**
 - For example, in AgF, Ag⁺ ($E^\ominus = +0.80$ V) and H⁺ ($E^\ominus = 0.00$ V) are present
 - Ag⁺ has the most positive $E^\ominus = +0.80$ V so is discharged, forming Ag
 - $\text{Ag}^+ + 2\text{e}^- \rightarrow \text{Ag}$
- The **negatively charged anion** with the most **negative** $E^\ominus$ will be discharged at the **anode**
 - For example, in FeSO₄, SO₄²⁻ and OH⁻ are present
 - $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ 4OH^- ($E^\ominus = -0.40$ V)
 - $\text{SO}_2 + 2\text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$ ($E^\ominus = -0.17$ V)
 - **Remember:** As the reverse reaction of that shown in Table 2.1 will occur, you must remember that the sign of $E^\ominus$ will be the opposite of the value given
 - OH⁻ has the most negative $E^\ominus$ so is discharged at the anode, forming O₂

2a

a) The value of $E^\ominus_{\text{cell}}$ under standard conditions and the negative electrode are:

- $E^\ominus_{\text{cell}} (= 0.80 - 0.77) = 0.03$ V

AND

Negative electrode: Pt / platinum **OR** left; [1 mark]

[Total: 1 mark]

- When calculating E^θ_{cell} subtract the less positive E^θ value from the more positive E^θ value
- In this case, the two equations, with their electrode potentials are:
 - $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag} \quad E^\theta = +0.80 \text{ V}$
 - $\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+} \quad E^\theta = +0.77 \text{ V}$
- So, $E^\theta_{\text{cell}} = 0.80 - 0.77 = 0.03 \text{ V}$
- The negative electrode is the electrode of the half-cell which has the less positive E^θ
- The $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell has the less positive E^θ so is the platinum electrode will be the negative electrode

2b

b) Comparing the actual E_{cell} of the above cell compare to the E^θ_{cell} under standard conditions:

- E_{cell} would be less positive / more negative; [1 mark]
- Because the concentration of $\text{Ag}^+ (\text{aq})$ is less than 1.0 mol dm^{-3} ; [1 mark]

[Total: 2 marks]

- Standard conditions require any ion concentrations to be 1.0 mol dm^{-3}
- As $\text{Ag}_2\text{SO}_4 (\text{s})$ is also present, some of the silver ions will form a solid with sulfate ions, removing them from the solution and causing the silver ion concentration to fall below 1.0 mol dm^{-3} , and the conditions are non-standard
- Consider the equilibrium:
 - $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$
- As the concentration of Ag^+ decreases, the equilibrium shifts to the left and the E value becomes less positive
- This causes E_{cell} to become less positive

2c

c)

i) The change in E_{cell} of the cell in Fig. 2.1 if a few cm^3 of concentrated $\text{Na}_2\text{SO}_4 (\text{aq})$ were added to the beaker containing $\text{Fe}^{3+} (\text{aq})$ and $\text{Fe}^{2+} (\text{aq})$ is:

- No change; [1 mark]

ii) The change in E_{cell} of the cell in Fig. 2.1 if a few cm^3 of concentrated Na_2SO_4 (aq) were added to the beaker containing Ag_2SO_4 (aq) is:

- More negative / less positive; [1 mark]

iii) The explanation for the change in E_{cell} is:

- The concentration of Ag^+ (aq) will decrease; [1 mark]
- Because $E_{\text{electrode}}$ becomes less positive

OR

Due to the common ion effect; [1 mark]

[Total: 4 marks]

- There is no change in (i) as the Na_2SO_4 dissolves into the solution and its ions have no effect on the equilibrium between Fe^{2+} and Fe^{3+}
- When Na_2SO_4 is added to the beaker containing Ag_2SO_4 , it does have an effect as they both contain the common ion, SO_4^{2-}
- The solution in the half-cell must be saturated as Ag_2SO_4 exists as both (aq) and (s)
- Therefore silver sulfate is in equilibrium with its ions:
 - $\text{Ag}_2\text{SO}_4 (\text{s}) \rightleftharpoons 2\text{Ag}^+ (\text{aq}) + \text{SO}_4^{2-} (\text{aq})$
- When Na_2SO_4 is added, the concentration of SO_4^{2-} ions increases, which shifts the above equilibrium to the left, causing more solid silver sulfate to form and reducing the concentration of Ag^+ ions in solution

b) Comparing the actual E_{cell} of the above cell compare to the E^{θ}_{cell} under standard conditions:

- E_{cell} would be less positive / more negative; [1 mark]
- Because the concentration of Ag^+ (aq) is less than 1.0 mol dm^{-3} ; [1 mark]

[Total: 2 marks]

- Standard conditions require any ion concentrations to be 1.0 mol dm^{-3}
- As Ag_2SO_4 (s) is also present, some of the silver ions will form a solid with sulfate ions, removing them from the solution and causing the silver ion concentration to fall below 1.0 mol dm^{-3} , and the conditions are non-standard
- Consider the equilibrium:
 - $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$
- As the concentration of Ag^+ decreases, the equilibrium shifts to the left and the E value becomes less positive
- This causes E_{cell} to become less positive

2c

c)

i) The change in E_{cell} of the cell in Fig. 2.1 if a few cm^3 of concentrated Na_2SO_4 (aq) were added to the beaker containing Fe^{3+} (aq) and Fe^{2+} (aq) is:

- No change; [1 mark]

ii) The change in E_{cell} of the cell in Fig. 2.1 if a few cm^3 of concentrated Na_2SO_4 (aq) were added to the beaker containing Ag_2SO_4 (aq) is:

- More negative / less positive; [1 mark]

iii) The explanation for the change in E_{cell} is:

- The concentration of Ag^+ (aq) will decrease; [1 mark]
- Because $E_{\text{electrode}}$ becomes less positive

OR

Due to the common ion effect; [1 mark]

[Total: 4 marks]

- There is no change in (i) as the Na_2SO_4 dissolves into the solution and its ions have no effect on the equilibrium between Fe^{2+} and Fe^{3+}
- When Na_2SO_4 is added to the beaker containing Ag_2SO_4 , it does have an effect as they both contain the common ion, SO_4^{2-}
- The solution in the half-cell must be saturated as Ag_2SO_4 exists as both (aq) and (s)
- Therefore silver sulfate is in equilibrium with its ions:
 - $\text{Ag}_2\text{SO}_4(\text{s}) \rightleftharpoons 2\text{Ag}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$
- When Na_2SO_4 is added, the concentration of SO_4^{2-} ions increases, which shifts the above equilibrium to the left, causing more solid silver sulfate to form and reducing the concentration of Ag^+ ions in solution
- Referring back to the equation at the electrode:
 - $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$
- As the concentration of Ag^+ decreases, this equilibrium shifts to the left, the E value becomes less positive and E_{cell} becomes less positive

2d

d) The equation to show the reaction taking place in the electrochemical cell in Fig. 2.1 is:

- $\text{Ag}^+ + \text{Fe}^{2+} \rightarrow \text{Ag} + \text{Fe}^{3+}$; [1 mark]

[Total: 1 mark]

- The two electrode reactions, with electrode potentials are:
 - $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag} \quad E^\theta = +0.80 \text{ V}$
 - $\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+} \quad E^\theta = +0.77 \text{ V}$
- As the first reaction has a more positive E value, Ag^+ is more readily reduced than Fe^{3+} as it accepts electrons more easily
- Fe^{2+} is more readily oxidised as it loses electrons more easily than Ag
- So the two half-equations are:
 - $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$
 - $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$
- The number of electrons in both half-equations is the same, so you can combine the two equations:
 - $\text{Ag}^+ + \text{e}^- + \text{Fe}^{2+} \rightarrow \text{Ag} + \text{Fe}^{3+} + \text{e}^-$
- Cancelling the electrons gives:
 - $\text{Ag}^+ + \text{Fe}^{2+} \rightarrow \text{Ag} + \text{Fe}^{3+}$

3a

a)

i) The overall cell reaction is:



ii) The voltage of this cell is:

- $E_{\text{cell}} (= 0.77 - 0.34) = 0.43 \text{ V}$; [1 mark]

[Total: 2 marks]

- As redox system **2** has a more positive E value, Fe^{3+} is more readily reduced than Cu^{2+} as it accepts electrons more easily
- Cu is more readily oxidised as it loses electrons more easily than Fe^{2+}
- So the two half-equations are:
 - $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$
 - $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
- The number of electrons in both half-equations is different, so multiply the bottom equation by 2 before combining the two equations:
 - $2\text{Fe}^{3+} + 2\text{e}^- + \text{Cu} \rightarrow 2\text{Fe}^{2+} + \text{Cu}^{2+} + 2\text{e}^-$
- Cancelling the electrons gives:
 - $2\text{Fe}^{3+} + \text{Cu} \rightarrow 2\text{Fe}^{2+} + \text{Cu}^{2+}$
- When calculating $E^\ominus_{\text{cell}}$ subtract the less positive $E^\ominus$ value from the more positive $E^\ominus$ value

3b

b)

i) The overall equations for **three** reactions that might be feasible are:

- $3\text{O}_2 + 12\text{H}^+ + 4\text{Al} \rightarrow 6\text{H}_2\text{O} + 4\text{Al}^{3+}$; [1 mark]
- $3\text{I}_2 + 2\text{Al} \rightarrow 2\text{Al}^{3+} + 6\text{I}^-$; [1 mark]
- $\text{O}_2 + 4\text{H}^+ + 4\text{I}^- \rightarrow 2\text{H}_2\text{O} + 2\text{I}_2$; [1 mark]

ii) **Two** reasons why these reactions might not take place, even if they are feasible are:

- The rate of reaction is slow

OR

E_a is high; [1 mark]

- The conditions are not standard / concentration is not 1.0 mol dm^{-3} ; [1 mark]

[Total: 5 marks]

- The possible reactions are between the three redox systems are **5 & 6**, **6 & 7** and **5 & 7**
- **A reaction is feasible between the forward reaction of the equation with the more positive E value and the reverse reaction of the equation with the more negative E value**
- Between **5 & 6**:
 - **5** has the more positive E value, so $\text{O}_2 + 4\text{H}^+$ have a greater tendency to gain electrons than Al^{3+} , so the half-equation is the forward reaction:
 - $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$
 - **6** has the more negative E value, so Al has a greater tendency to lose electrons than H_2O , so the half-equation is the reverse reaction:
 - $\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$
 - **Remember:** When combining the half-equations, the number of electrons in each half-equation must be the same so that the electrons cancel
 - Multiplying the two half-equations so that they both have 12 electrons:
 - $3\text{O}_2 + 12\text{H}^+ + 12\text{e}^- \rightarrow 6\text{H}_2\text{O}$
 - $4\text{Al} \rightarrow 4\text{Al}^{3+} + 12\text{e}^-$
 - Combining the half-equations gives:
 - $3\text{O}_2 + 12\text{H}^+ + 4\text{Al} \rightarrow 6\text{H}_2\text{O} + 4\text{Al}^{3+}$
- Between **6 & 7**:
 - **7** has the more positive E value, so I_2 has a greater tendency to gain electrons than Al^{3+} , so the half-equation is the forward reaction:
 - $\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$
 - **6** has the more negative E value, so Al has a greater tendency to lose electrons than I^- , so the half-equation is the reverse reaction:
 - $\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$
 - Multiplying the two half-equations so that they both have 6 electrons:
 - $3\text{I}_2 + 6\text{e}^- \rightarrow 6\text{I}^-$
 - $2\text{Al} \rightarrow 2\text{Al}^{3+} + 6\text{e}^-$
 - Combining the half-equations gives:
 - $3\text{I}_2 + 2\text{Al} \rightarrow 2\text{Al}^{3+} + 6\text{I}^-$

- Between **5** & **7**:
 - **5** has the more positive E value, so $\text{O}_2 + 4\text{H}^+$ have a greater tendency to gain electrons than I_2 , so the half-equation is the forward reaction:
 - $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$
 - **7** has the more negative E value, so I^- has a greater tendency to lose electrons than H_2O , so the half-equation is the reverse reaction:
 - $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$
 - Multiplying the second half-equation so that they both have 4 electrons:
 - $4\text{I}^- \rightarrow 2\text{I}_2 + 4\text{e}^-$
 - Combining the half-equations gives:
 - $\text{O}_2 + 4\text{H}^+ + 4\text{I}^- \rightarrow 2\text{H}_2\text{O} + 2\text{I}_2$

3c

c)

i) An oxidising agent that oxidises $\text{Fe}^{2+}(\text{aq})$ to $\text{Fe}^{3+}(\text{aq})$ is:

- Cl_2
OR
 O_2 **AND** H^+ ; [1 mark]

ii) A species that reduces $\text{Fe}^{3+}(\text{aq})$ to Fe^{2+} but does not reduce $\text{Cu}^{2+}(\text{aq})$ to $\text{Cu}(\text{s})$ is:

- I^- ; [1 mark]

[Total: 2 marks]

- For Fe^{2+} to be oxidised to Fe^{3+} , the reverse reaction in redox system 6 must be favoured
- This means that you need to find a redox system that has a more positive E^θ , which means the forward reaction will be favoured
- Redox system 4 has a more positive E^θ value, so Cl_2 is more easily reduced and will act as an oxidising agent
- As the $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox system 2 has a less positive E^θ , Fe^{2+} is more easily oxidised
- Similarly, redox system 5 has a more positive E^θ value, so O_2 and H^+ will oxidise Fe^{2+} and you must give both of these species to gain this mark
- For Fe^{3+} to be reduced to Fe^{2+} you need to find a redox system that has an E^θ value that is less positive, i.e. less than +0.77 V

- However, it needs to be more positive than E^θ of redox system **3**, i.e. +0.34 V, otherwise it would also reduce Cu^{2+} ions
- The redox system with a value between these is **7** which means that I^- can reduce Fe^{3+}
- **Careful:** Reducing agents are on the right, so you would not gain a mark if you gave I_2 as the species that reduces Fe^{2+}

3d

d) The electrode potential at 298 K of redox system **3** if the concentration of Cu^{2+} (aq) ions was $0.0002 \text{ mol dm}^{-3}$ is:

- $E = E^\theta + \frac{0.059}{z} \log_{10} \frac{[\text{oxidised species}]}{[\text{reduced species}]}$; [1 mark]
- $E = 0.34 + \frac{0.059}{2} \log_{10}(0.0002)$; [1 mark]
- $E (= 0.34 - 0.11) = 0.23 / 0.231 / 0.2309 \text{ V}$; [1 mark]

[Total: 3 marks]

- You are expected to know the Nernst equation in the form given in the first marking point, which is only applicable at 298 K
- The reduced species is the metal, Cu – the concentration of the metal does not alter so the term $\frac{[\text{oxidised species}]}{[\text{reduced species}]}$ can be rewritten as $[\text{Cu}^{2+}(\text{aq})]$

act as an oxidising agent

- As the $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox system **2** has a less positive $E^\ominus$, Fe^{2+} is more easily oxidised
- Similarly, redox system **5** has a more positive $E^\ominus$ value, so O_2 and H^+ will oxidise Fe^{2+} and you must give both of these species to gain this mark
- For Fe^{3+} to be reduced to Fe^{2+} you need to find a redox system that has an $E^\ominus$ value that is less positive, i.e. less than +0.77 V
- However, it needs to be more positive than $E^\ominus$ of redox system **3**, i.e. +0.34 V, otherwise it would also reduce Cu^{2+} ions
- The redox system with a value between these is **7** which means that I^- can reduce Fe^{3+}
- **Careful:** Reducing agents are on the right, so you would not gain a mark if you gave I_2 as the species that reduces Fe^{2+}

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- $E = 0.34 + \frac{0.059}{2} \log_{10}(0.0002)$; [1 mark]
- $E (= 0.34 - 0.11) = 0.23 / 0.231 / 0.2309 \text{ V}$; [1 mark]

[Total: 3 marks]

- You are expected to know the Nernst equation in the form given in the first marking point, which is only applicable at 298 K
- The reduced species is the metal, Cu – the concentration of the metal does not alter so the term $\frac{[\text{oxidised species}]}{[\text{reduced species}]}$ can be rewritten as $[\text{Cu}^{2+}(\text{aq})]$
- z refers to the number of electrons transferred in the reaction – in this equation there are 2 electrons transferred, so $z = 2$
- Then it is a matter of substituting the values into the equation
- As the concentration of $\text{Cu}^{2+}(\text{aq})$ is less than 1.0 mol dm^{-3} , $\log_{10}[\text{Cu}^{2+}]$ is negative and E is less positive than $E^\ominus$

4a

a) To identify the weakest oxidising agent:

- $\text{Fe}^{2+} / \text{Fe(II)} / \text{iron(II) ion}$; [1 mark]
- It has the smallest / most negative electrode potential / $E^\ominus$; [1 mark]

[Total: 2 marks]

- The weakest oxidising agent (strongest reducing agent) loses electrons easily and has the most negative $E^\ominus$ value
- Look at the values in the table, the most negative $E^\ominus$ value is the weakest oxidising agent
 - To explain your choice you need to explain that you have chosen the most negative value
- Redox equations with more negative potentials correspond to electron loss and so go from right to left
- **Careful:** You would not gain the first mark for just identifying which redox system has the most negative $E^\ominus$, i.e. $\text{Fe} / \text{Fe}^{2+}$, you must specifically identify the iron(II) ion as the weakest oxidising agent
- You can only score the second mark if you gain the first mark

4b

b) The standard cell potential of this cell is:

- $E^\ominus_{\text{cell}} = +0.34\text{V} - (-0.25\text{V}) = 0.59\text{V}$; [1 mark]

[Total: 1 mark]

- To calculate the standard cell potential, subtract the less positive $E^\ominus$ from the more positive $E^\ominus$
 - The more negative electrode potential is the half-equation that is being oxidised, and the reverse reaction occurs:
 - $\text{Ni (s)} \rightarrow \text{Ni}^{2+} (\text{aq}) + 2\text{e}^-$
 - The more positive electrode potential is the half equation that is being reduced, and the forward reaction occurs:
 - $\text{Cu}^{2+} (\text{aq}) + 2\text{e}^- \rightarrow \text{Cu (s)}$
 - The overall equation is:
 - $\text{Ni (s)} + \text{Cu}^{2+} (\text{aq}) \rightarrow \text{Ni}^{2+} (\text{aq}) + \text{Cu (s)}$

4c

c) The standard Gibbs free energy change for the electrochemical cell in part (b) is:

- $\Delta G^\theta = -n \times E^\theta_{\text{cell}} \times F$; [1 mark]
- $\Delta G^\theta = -2 \times 0.59 \times 9.65 \times 10^4$; [1 mark]
- $\Delta G^\theta = -113870 / -114000$ **AND** J mol^{-1}
OR
 $-113.87 / -114$ **AND** kJ mol^{-1} ; [1 mark]
- The reaction is feasible; [1 mark]

[Total: 4 marks]

- n represents the number of electrons transferred in the cell reaction
 - Ni loses 2 electrons and Cu^{2+} gains 2 electrons, so $n = 2$
- E^θ_{cell} is the value you calculated in part (b), you would be allowed error carried forward if you got this value incorrect
- The question gives you F
- Substitute the values into the equation to calculate ΔG^θ
- **Careful:** The units of ΔG^θ as calculated by the equation are J mol^{-1} , you can leave the answer like this or convert to kJ mol^{-1}
- Your calculated answer should be given to 3 significant figures or more
- As ΔG^θ is negative, the reaction is feasible

4d

d)

i) The two half equations that are connected in the cell are:

- Fe/Fe^{2+} **AND** Sn^{2+}/Sn ; [1 mark]

ii) The half-equation for the reaction that occurs at the cathode is:

- $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$; [1 mark]

[Total: 2 marks]

- First, to deduce which two half-cells are involved the values for the electrode potentials must be considered

- Only the Fe^{2+} and Sn^{2+} electrode potentials have a difference of 0.31 V
- $E^\ominus_{\text{cell}} = -0.14 - (-0.45) = 0.31 \text{ V}$
- In this question the iron electrode potential is more negative than the tin electrode potential, so will be the negative pole
- The half-equation with the more negative electrode will favour the reverse reaction as it is more easily oxidised
 - The half-equation at the cathode is:
 - $\text{Fe}(\text{s}) \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$
- The half-equation with the more positive electrode will favour the forward reaction as it is more easily reduced, so will be the positive pole
 - The half-equation at the anode is:
 - $\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$
- The overall cell equation is:
 - $\text{Fe}(\text{s}) + \text{Ni}^{2+}(\text{aq}) \rightarrow \text{Fe}^{2+}(\text{aq}) + \text{Ni}(\text{s})$

5a

a) Equations for the feasible reactions are:

- $3\text{Ag}^+ + \text{Cr} \rightarrow 3\text{Ag} + \text{Cr}^{3+}$; [1 mark]
- $2\text{Cr}^{3+} + 3\text{Mg} \rightarrow 2\text{Cr} + 3\text{Mg}^{2+}$; [1 mark]
- $2\text{Ag}^+ + \text{Mg} \rightarrow 2\text{Ag} + \text{Mg}^{2+}$; [1 mark]

[Total: 3 marks]

- The possible reactions are between redox systems **1 & 2**, **2 & 3** and **1 & 3**
- **A reaction is feasible between the forward reaction of the equation with the more positive *E* value and the reverse reaction of the equation with the more negative *E* value**
- Between **1 & 2**:
 - **1** has the more positive *E* value, so Ag^+ has a greater tendency to gain electrons than Cr^{3+} , so the half-equation is the forward reaction:
 - $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$
 - **2** has the more negative *E* value, so Cr has a greater tendency to lose electrons than Ag, so the half-equation is the reverse reaction:
 - $\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$
 - **Remember:** When combining the half-equations, the number of electrons in each half-equation must be the same so that the electrons cancel
 - Multiplying the Ag^+/Ag half-equation by 3 to give the same number of

electrons as the Cr^{3+}/Cr half-equation and combining gives:



- Between **2** & **3**:

- **2** has the more positive E value, so Cr^{3+} has a greater tendency to gain electrons than Mg^{2+} , so the half-equation is the forward reaction:



- **3** has the more negative E value, so Mg has a greater tendency to lose electrons than Cr, so the half-equation is the reverse reaction:



- Multiplying the Cr^{3+}/Cr half-equation by 2 and the Mg^{2+}/Mg half-equation by 3 to give 6 electrons in both half-equations and combining gives:



- Between **1** & **3**:

- **1** has the more positive E value, so Ag^+ has a greater tendency to gain electrons than Mg^{2+} , so the half-equation is the forward reaction:



- **3** has the more negative E value, so Mg has a greater tendency to lose electrons than Ag, so the half-equation is the reverse reaction:



- Multiplying the Ag^+/Ag half-equation by 2 to give the same number of electrons as the Mg^{2+}/Mg half-equation and combining gives:



- **2** has the more negative *E* value, so Cr has a greater tendency to lose electrons than Ag, so the half-equation is the reverse reaction:
 - $\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^{-}$
- **Remember:** When combining the half-equations, the number of electrons in each half-equation must be the same so that the electrons cancel
- Multiplying the Ag^{+}/Ag half-equation by 3 to give the same number of electrons as the Cr^{3+}/Cr half-equation and combining gives:
 - $3\text{Ag}^{+} + \text{Cr} \rightarrow 3\text{Ag} + \text{Cr}^{3+}$
- Between **2** & **3**:
 - **2** has the more positive *E* value, so Cr^{3+} has a greater tendency to gain electrons than Mg^{2+} , so the half-equation is the forward reaction:
 - $\text{Cr}^{3+} + 3\text{e}^{-} \rightarrow \text{Cr}$
 - **3** has the more negative *E* value, so Mg has a greater tendency to lose electrons than Cr, so the half-equation is the reverse reaction:
 - $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^{-}$
 - Multiplying the Cr^{3+}/Cr half-equation by 2 and the Mg^{2+}/Mg half-equation by 3 to give 6 electrons in both half-equations and combining gives:
 - $2\text{Cr}^{3+} + 3\text{Mg} \rightarrow 2\text{Cr} + 3\text{Mg}^{2+}$
- Between **1** & **3**:
 - **1** has the more positive *E* value, so Ag^{+} has a greater tendency to gain electrons than Mg^{2+} , so the half-equation is the forward reaction:
 - $\text{Ag}^{+} + \text{e}^{-} \rightarrow \text{Ag}$
 - **3** has the more negative *E* value, so Mg has a greater tendency to lose electrons than Ag, so the half-equation is the reverse reaction:
 - $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^{-}$
 - Multiplying the Ag^{+}/Ag half-equation by 2 to give the same number of electrons as the Mg^{2+}/Mg half-equation and combining gives:
 - $2\text{Ag}^{+} + \text{Mg} \rightarrow 2\text{Ag} + \text{Mg}^{2+}$

5b

b) The standard cell potential is:

- $E^{\ominus}_{\text{cell}} (= -0.74 - (-2.37) = (+)1.63 \text{ V}; [1 \text{ mark}]$

ii) The sign of the electrode in redox system **2** of the cell is:

- Positive; [1 mark]

[Total: 2 marks]

- $E^\ominus_{\text{cell}}$ can be calculated by subtracting the less positive $E^\ominus$ from the more positive $E^\ominus$ value
- **Careful:** Here there are two negative numbers, -0.74 is more positive than -2.37 , so -2.37 is subtracted from -0.74 :
 - $-0.74 - (-2.74) = +1.63 \text{ V}$
 - You do not need to show the positive sign for this first mark
- Redox system **2** has the more positive $E^\ominus$ value, so will be the positive pole and so the electrode will be positive

5c

c) The cell potential will:

- Be more positive; [1 mark]
- Because the concentration of Mg^{2+} ions decreases; [1 mark]
- $E^\ominus$ is more negative; [1 mark]

[Total: 3 marks]

- Diluting the solution in redox system **3** with distilled water will decrease the concentration of $\text{Mg}^{2+}(\text{aq})$ ions
- Looking at the equation:
 - $\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Mg}(\text{s}) \quad E^\ominus = -2.37 \text{ V}$
- The position of equilibrium will shift to the left
 - This means Mg^{2+} will be less easily reduced as it does not gain electrons as easily
 - The $E^\ominus$ value will become more negative
- This means that E_{cell} will become more positive
- It can be difficult to see what happens to E_{cell} as the $E^\ominus$ value changes, try using some numbers to help:
 - As the $E^\ominus$ value for redox system **3** becomes more negative, choose a number that is more negative than -2.37 V , e.g. -3.0 V
 - Now, use this more negative number to calculate E_{cell} :
 - $E_{\text{cell}} = -0.74 - (-3.00) = +2.26 \text{ V}$
 - The original $E^\ominus_{\text{cell}}$ was $+1.63 \text{ V}$, so the new E_{cell} is **more positive** than the original $E^\ominus_{\text{cell}}$

5d

d) The new electrode potential, E , for redox system **2** would be:

- $E = -0.74 + \frac{0.059}{3} \log_{10}(0.01)$; [1 mark]
- $E = -0.78 \text{ V} / -0.7793$; [1 mark]

[Total: 2 marks]

- This form of the Nernst equation is valid at 298 K
- z is the number of electrons transferred - use the half-equation you can see that 3 electrons are transferred
- The term $\frac{[\text{oxidised species}]}{[\text{reduced species}]}$ can be written as [oxidised species]. i.e. $[\text{Cr}^{3+}(\text{aq})] = 0.01 \text{ mol dm}^{-3}$
 - This is because the concentration of the metal does not change
- It is then just a matter of substituting the values into the equation given
- **Note:** If the concentration of the solution is less than 1 mol dm^{-3} , you would expect E to be less positive than E^θ , so make sure that your answer is less positive than -0.74 V

